

QUANTITATIVE TRADING × ADVERSARIAL ML

QuantAdv

An Adversarial Machine Learning Sandbox for Safe, Realistic
Algorithmic Trading Research



Simulated markets



Attack simulation



Defensive AI

OVERVIEW

What QuantAdv Actually Does

QuantAdv is a full-stack simulation platform that fuses quantitative finance, machine learning, and cybersecurity. It lets users build and test AI trading strategies on real market data — with zero real money at risk — while also studying how those same AI models can be attacked and defended.



Trade

Predict prices & run paper-trading strategies



Attack

Launch adversarial attacks on the trading models



Defend

Harden models and measure their resilience

WHY IT MATTERS

AI now drives real investment decisions — but AI models can be fooled.

A single manipulated data point can shift a trading signal from Buy to Sell — QuantAdv is the safe lab to see exactly how, and how to stop it.

Built for students, researchers & FinTech teams to learn AI security hands-on — without touching a live market.

PROBLEM STATEMENT

AI Trading Systems Are Being Trusted Faster Than They're Being Secured

Modern financial markets increasingly rely on AI-driven systems to make real-time investment decisions. But these models are vulnerable to adversarial machine learning attacks — where bad actors manipulate input data or poison training sets to bend predictions in their favor, risking real financial loss.



The gap

There's no safe, hands-on place to study these vulnerabilities, test trading algorithms, and evaluate defenses — without risking real capital.

Zero

REAL CAPITAL AT RISK

4+

ADVERSARIAL ATTACK TYPES SIMULATED

OBJECTIVES

What The Project Sets Out To Do

01



Predict market trends

Develop AI models that forecast price movement across stocks, crypto & forex.

02



Simulate trading strategies

Run algorithmic strategies in a realistic virtual market environment.

03



Demonstrate adversarial attacks

Show concretely how manipulated data can fool ML trading models.

04



Build defense mechanisms

Apply robust ML techniques that harden models against those attacks.

05



Visualize performance & risk

Surface trading and security metrics through interactive dashboards.

KEY FEATURES

Ten Capabilities In One Sandbox



Real-time market data

Stocks, crypto & forex feeds



AI price prediction

LSTM, Transformer & tree models



Paper trading simulator

Risk-free strategy execution



Strategy backtesting

Quantitative performance testing



Adversarial attack simulation

FGSM, PGD, poisoning & more



Defense mechanism testing

Adversarial training & distillation



Portfolio & risk analytics

Live growth & exposure tracking



Explainable AI (XAI)

SHAP-based model transparency



Performance reports

Exportable trading + security metrics



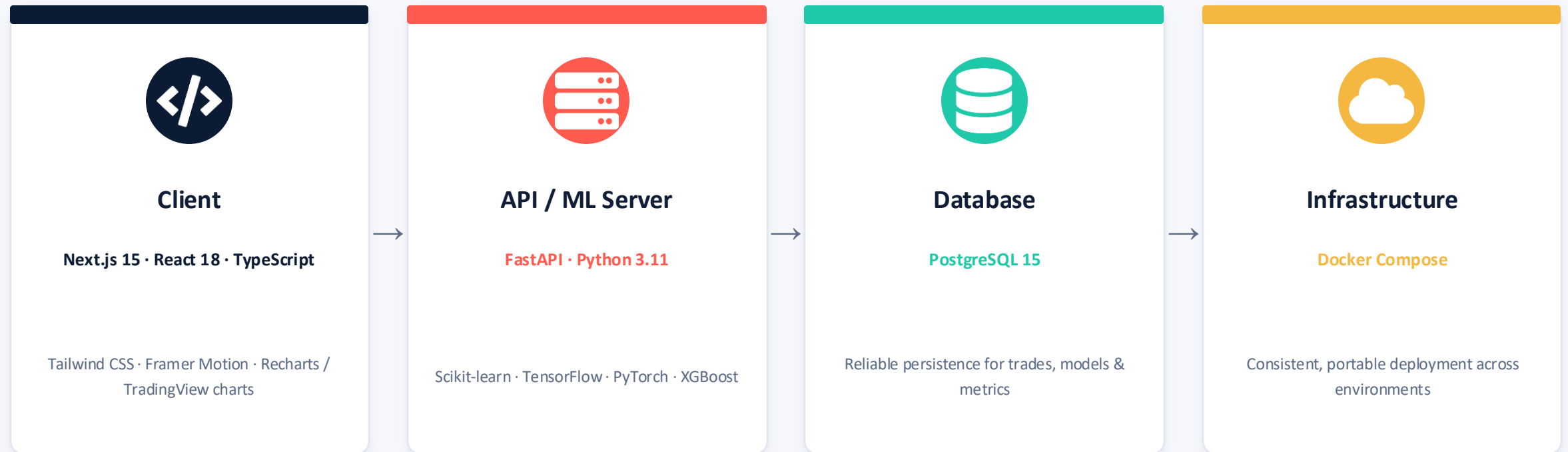
Admin dashboard

Full platform oversight & control

FRAMEWORK

System Architecture

A modern, containerized full-stack built for real-time simulation at scale.



Deployment: frontend on Vercel · backend via Gunicorn + Uvicorn · managed Postgres (e.g. Supabase / RDS)

WORKFLOW

How A Session Runs, Start To Finish



Steps 6 & 7 are the sandbox's core loop — attack, then defend — repeated to continuously harden the model.

Attack The Model, Then Armor It



ADVERSARIAL ATTACKS

- **FGSM**

Fast Gradient Sign Method — quick, single-step perturbations

- **PGD**

Projected Gradient Descent — stronger, iterative perturbations

- **Data Poisoning**

Corrupting training data to bias the model

- **Label Flipping**

Mislabeling training samples to mislead learning



DEFENSIVE MECHANISMS

- **Adversarial Training**

Training on attack examples so the model learns to resist them

- **Defensive Distillation**

Smoothing model outputs to blunt gradient-based attacks

- **Outlier Detection**

Flagging suspicious/manipulated inputs before they're used

- **Explainable AI (SHAP)**

Making model decisions inspectable, so tampering is visible

APPLICATIONS

Who This Is Useful For



Algorithmic traders

Prototype & backtest strategies without financial risk



AI security researchers

Study real adversarial attack/defense trade-offs



Students & universities

Hands-on hackathon and coursework projects



FinTech product teams

Prototype AI trading features before production



ML researchers

Benchmark model robustness under attack



Risk & compliance teams

Understand where AI-driven decisions can fail

EXPECTED OUTCOMES

What Success Looks Like



Accurate, AI-based market predictions across asset classes



ML models measurably more resistant to adversarial attacks



Clear, practical understanding of AI security in finance



Rigorous strategy evaluation through realistic paper trading



Interactive analytics for portfolio growth, accuracy & attack success rates

Where QuantAdv Goes Next

- Reinforcement learning-based trading agents
- Live paper trading with real-time market feeds
- Sentiment analysis from financial news & social media
- Blockchain-based trade verification
- Federated learning for secure collaborative AI
- Multi-agent AI trading simulations
- Deep-RL-driven automated portfolio optimization



CONCLUSION

One Sandbox. Finance, AI & Security — Together.

QuantAdv brings Finance, AI, Machine Learning, Data Analytics, and Cybersecurity into a single platform — letting users safely experiment with trading algorithms, understand real AI vulnerabilities, and build robust defenses. It's a practical, hands-on foundation for the secure, intelligent financial systems the industry now needs.

[Explore the project](#)

quantadv.vercel.app

github.com/John-Varghese-EH/Quantitative-Trading

Thank you